Lecture 6

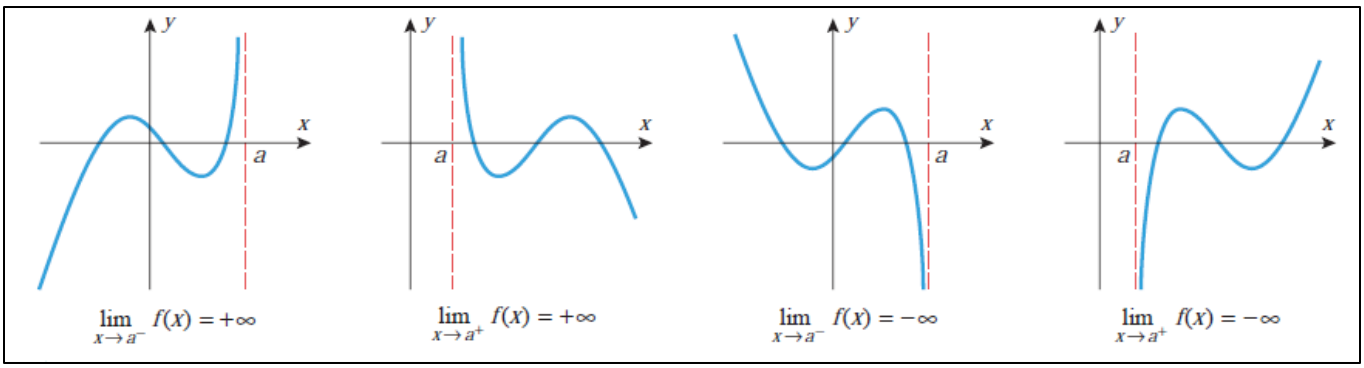
Asymptotes

Vertical Asymptotes: In infinite limits,

$$\lim_{x\to a^-} f(x) = +\infty \qquad \lim_{x\to a^+} f(x) = +\infty$$

$$\lim_{x\to a^-} f(x) = -\infty \qquad \lim_{x\to +} f(x) = -\infty$$

In each case, $y = f(x)$ either rises or falls without bounds, squeezing closer and closer to the vertical line $x = a$ as x approaches a from the side indicated in the limit. The line $x = a$ is called a vertical asymptote of the curve $y = f(x)$.

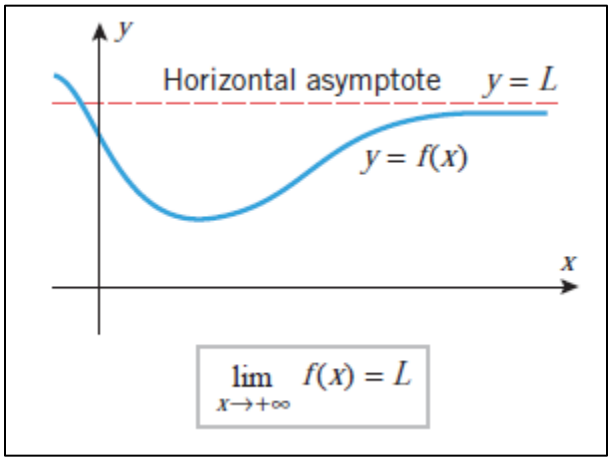


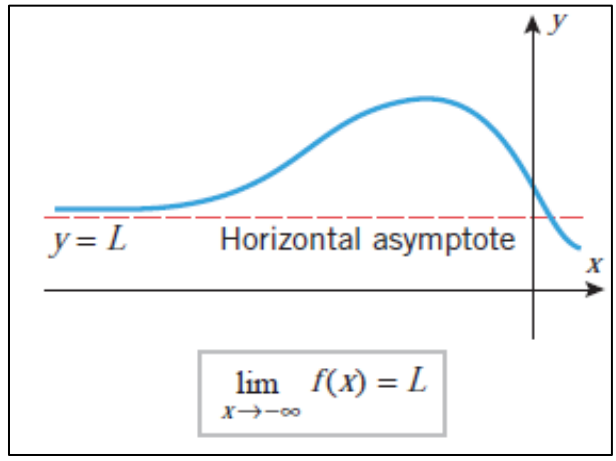
Horizontal Asymptotes:

$$\lim_{x\to +\infty} f(x) = L \qquad \lim_{x\to -\infty} f(x) = L$$

In the first case, the graph of f eventually comes as close as we like to the line $y = L$ as x increases without bound, and in the second case it eventually comes as close as we like to the line $y = L$ as x decreases without bound. If either limit holds, we call the line $y = L$ a horizontal asymptote for the graph of f.

Then, $y = L$ is the horizontal asymptote.





Example: $f(x) = \frac{x^2+6x+8}{x^2+3x-4}$

find the discontinuities, holes, vertical and horizontal asymptotes, x and y-intercepts, and domain. Then sketch the function.

Solution: $f(x) = \frac{x^2+6x+8}{x^2+3x-4} = \frac{(x+4)(x+2)}{(x+4)(x-1)} = \frac{x+2}{x-1}$

<u>Discontinuities:</u> $x + 4 = 0 \Rightarrow x = -4$

$x - 1 = 0 \Rightarrow x = 1$

<u>Holes:</u> (removable discontinuity)

$x + 4 = 0$

$\Rightarrow x = -4$

<u>Vertical Asymptote:</u> (infinite discontinuity)

$x - 1 = 0$

$\Rightarrow x = 1$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} \frac{x+2}{x-1} = +\infty$$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \frac{x+2}{x-1} = -\infty$$

<u>Horizontal Asymptote:</u>

$$\lim_{x\to -\infty} \frac{x+2}{x-1} = \lim_{x\to -\infty} \frac{1+\frac{2}{x}}{1-\frac{1}{x}} = \frac{1}{1} = 1$$

$$\lim_{x\to+\infty}\frac{x+2}{x-1}=\lim_{x\to+\infty}\frac{1+\frac{2}{x}}{1-\frac{1}{x}}=\frac{1+0}{1-0}=1$$

So, $y=1$.

<u>x-intercept:</u>

$$y=0=\frac{x+2}{x-1}$$

$\Rightarrow 0=x+2$

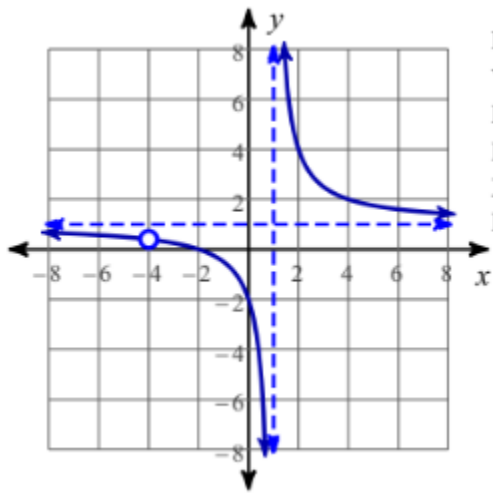
$\Rightarrow x=-2$

<u>y-intercept:</u>

$$y=\frac{0+2}{0-1}=-2$$

<u>Domain:</u>

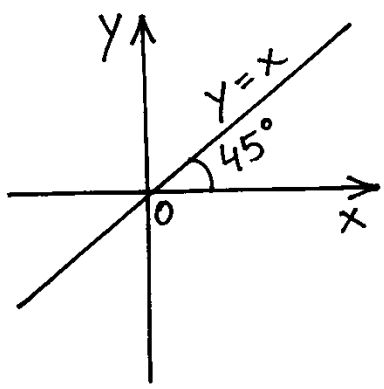
$\mathbb{R}\backslash\{-4,1\}$



<u>H.W. PRACTICE SHEET</u>

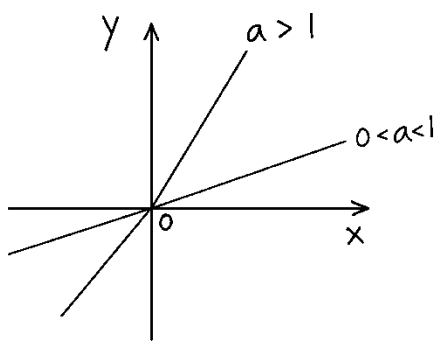
<u>Sketch:</u>

1. $f(x)=x$
   $y=x$



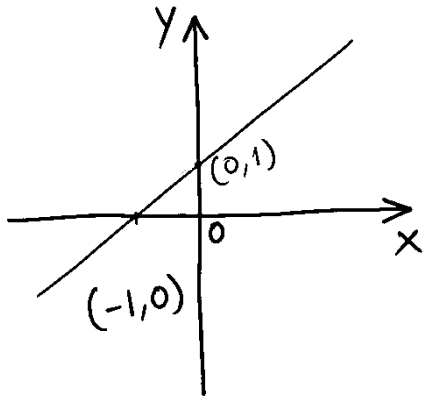
2. $f(x)=ax$

   $y=ax$



3. $f(x) = y = mx + c$
   $f(x) = x + 1$
   $y = x + 1$
   $(y - 1) = x$

   Here, if $y = 0, x = -1$.
   if $x = 0, y = 1$.

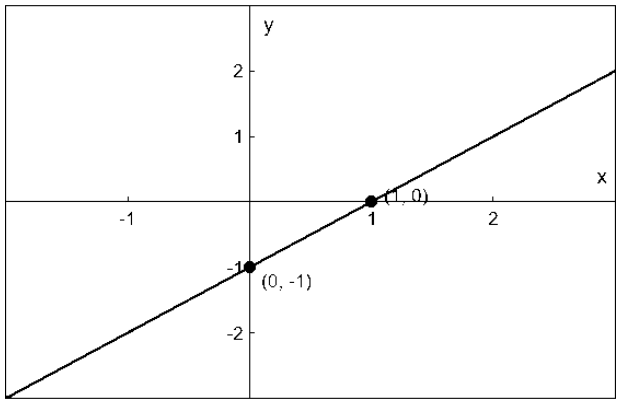


4. $f(x) = x - 1$

   $y = x - 1$

   Here, if $y = 0, x = 1$.

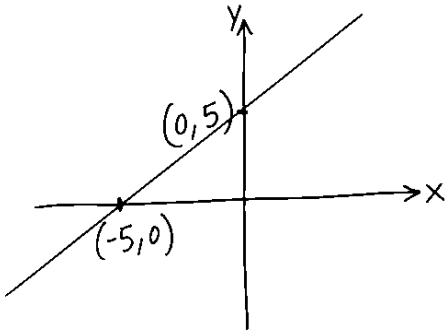
   if $x = 0, y = -1$.



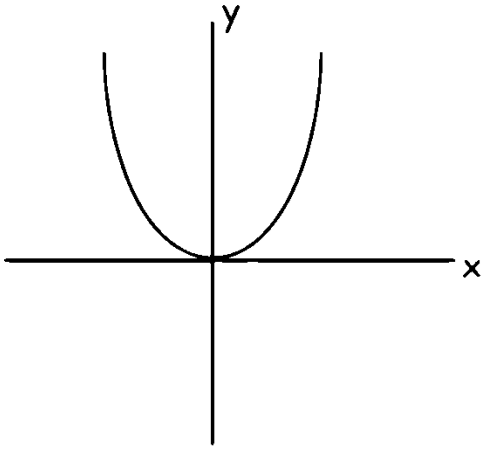
5. $y - 2 = x + 3$
   $y = x$+3+2
   $y = x$+5

   Here, if $y = 0, x = -5$.

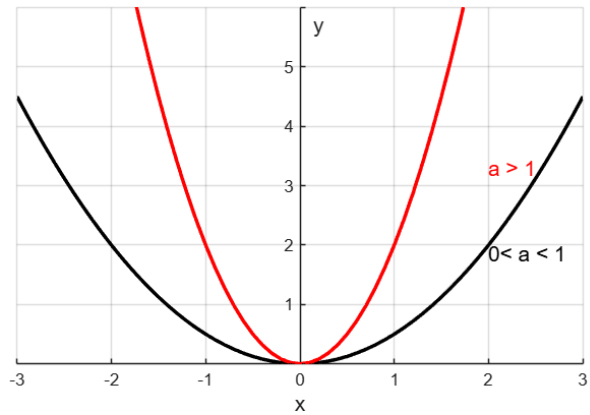
   if $x = 0, y = 5$.



6. $y = x^2$

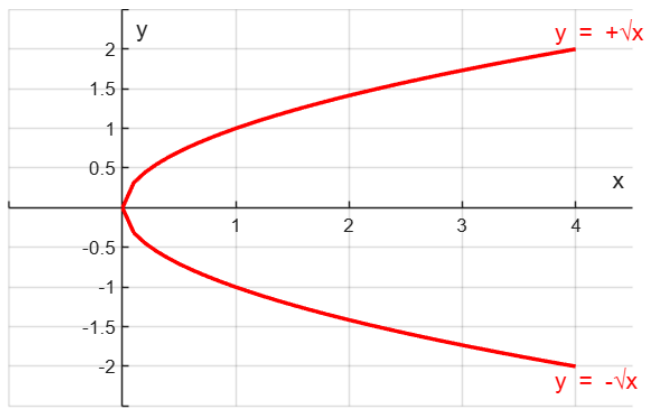


7. $y = ax^2$



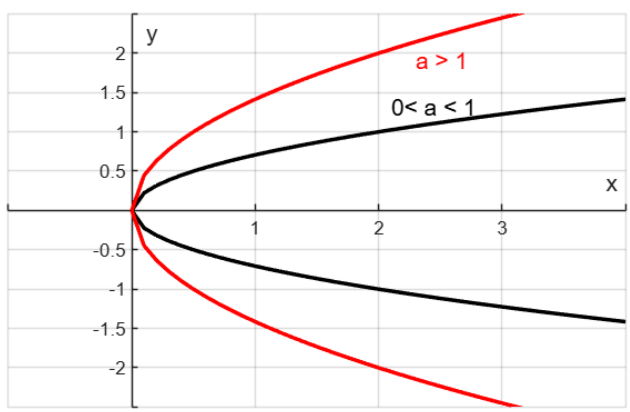
8. $y^2 = x$

$y = \pm\sqrt{x}$



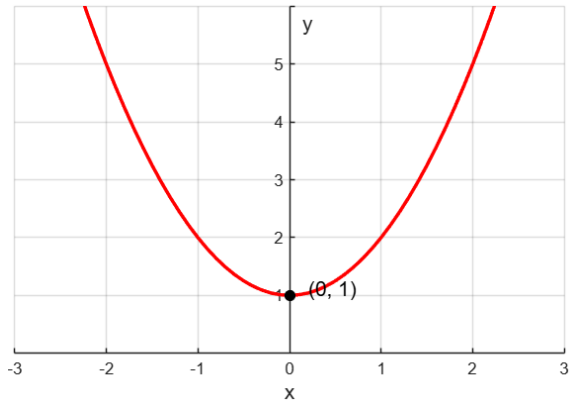
9. $y^2 = ax$

$y = \pm\sqrt{ax}$



10. $y = x^2 + 1$

$x^2 = y - 1$



11. $y = (x + 1)^2$

Here, if $y = 0$

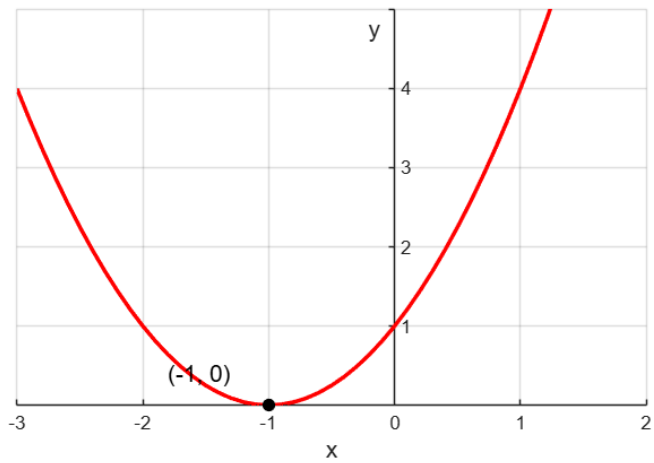
$(x + 1)^2 = 0$

$x + 1 = 0$

$x = -1$

Here, if $x = 0$

$y = 1$



12. $y - 2 = (x + 1)^2$

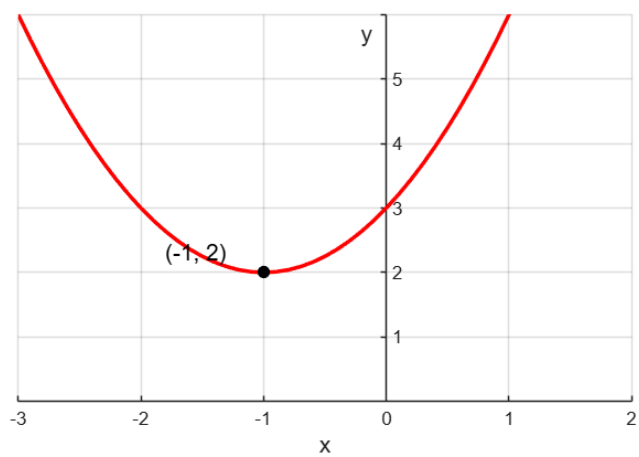
Here, if $y = 0$

$(x + 1)^2 = -2$

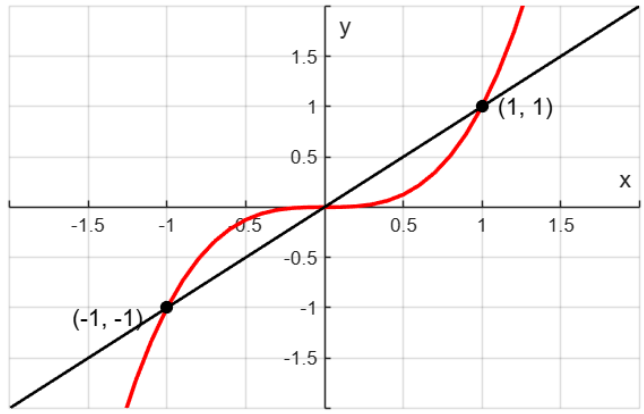
Which is not possible.

Here, if $x = 0$

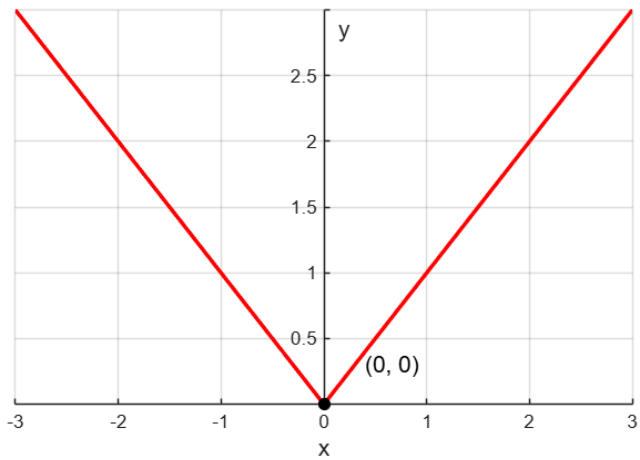
$y = 3$



13. $y = x^3$



14. $y = |x|$



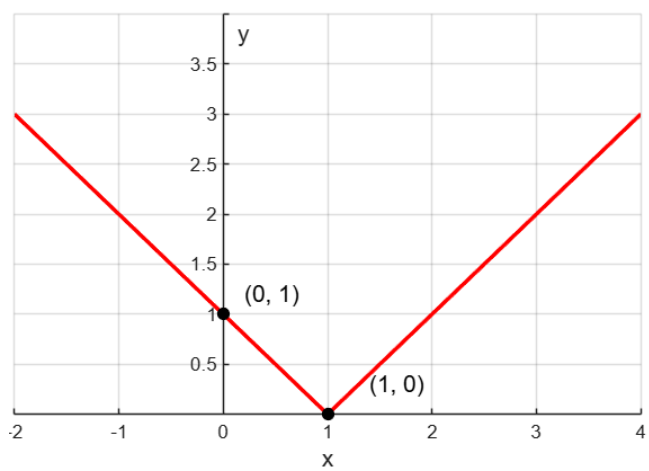
15. $y = |x - 1|$

Here, if $y = 0$

$x = 1$

Here, if $x = 0$

$y = 1$



16. $y = |x - 1| + 3$
Or, $y - 3 = |x - 1|$

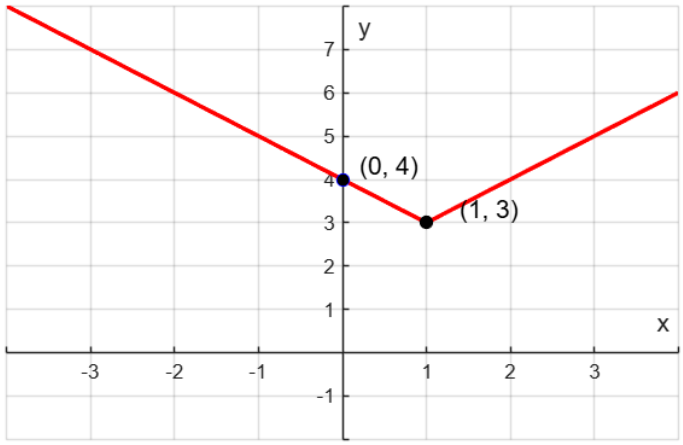
Here, if $y = 0$

$$|x - 1| = -3$$

Which is not possible.

Here, if $x = 0$

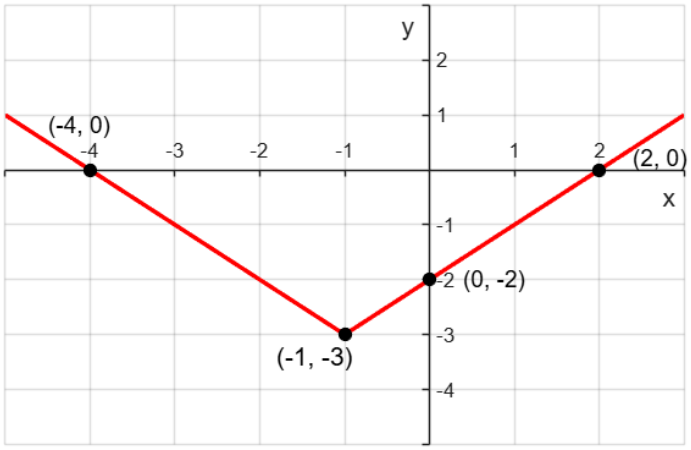
$$y = 4$$



17. $y = |x + 1| - 3$
Or, $y + 3 = |x + 1|$

Here, if $y = 0$

$$x = 2, -4$$

Here, if $x = 0$

$$y = -2$$